

VERTICAL STRIP PLUNGER EJECTION REPORT

PROJECT BLISTER BOT — VERTICAL 2-AXIS CNC & PLUNGER DYNAMICS

Module: Vertical Scoring Mechanism	Strip Orientation: Vertical (Cartridge)	Plunger Action: Horizontal Push
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1. Mechanical Innovation: Vertical Strip & Horizontal Plunger Ejection

In **Blister Bot**, medicine blister strips are held vertically inside multi-slot cartridges. The deblistering toolhead operates on a vertical 2-axis CNC gantry behind the vertical foil sheet. The motorized rotary cutter wheel scores a 270-degree U-shaped flap in the vertical foil card. A soft silicone plunger pin then extends horizontally (+Z direction) through the vertical strip, pushing the capsule tablet out of the front PVC bubble so it drops into the slanted delivery chute below.

2. Rendered 3D Kinematic Previews



3. Kinematic Sequence Parameters

- **Step 1 (Vertical Align):** Gantry positions toolhead carriage vertically (Y) and horizontally (X) over target bubble.
- **Step 2 (Blade Engage):** Rotary cutter wheel engages forward onto vertical foil backing.
- **Step 3 (270° U-Scoring):** Cutter traverses 3-sided U-path, leaving top edge uncut to form hinged aluminum flap.
- **Step 4 (Open Foil Flap):** Cutter retracts; vertical aluminum flap swings open 75°.
- **Step 5 (Horizontal Plunger Eject):** Silicone plunger pin extends horizontally 14mm, pushing capsule tablet out into slanted chute.

4. Vertical Scoring Folder Inventory (vertical scoring mechanism/)

File Name	Function / Purpose
vertical_strip_plunger_animation.html	Interactive 3D WebGL Three.js animation web app for vertical strip ejection.
vertical_strip_ejection_demo.mp4	Generated MP4 animation video showing vertical scoring & plunger ejection.
vertical_scoring_mechanism.blend	Native 3D Blender project model file (vertical strip orientation).
animate_vertical_scoring.py	Blender Python script for 3D model generation & keyframe rendering.
encode_vertical_mp4.py	OpenCV Python script encoding PNG frames to MP4 video format.
Vertical_Scoring_Report.pdf	Publication-ready PDF documentation report.